DUE 9/10/26
COSC559

Mini-Project 1

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Question

Does Jane Austen’s writing style and vocabulary change between her debut novel *Sense and Sensibility* and her later *Pride and Prejudice* novel? Did she become “bolder” and more confident in her writing as she gained experience?

How does Susan Ferrier *Marriage* (1818), written by an author who is known as a “Scottish Austen”, compare to Austen’s own language from roughly the same time period? Does Ferrier’s Scottish identity show up differently in her works than Austen’s English social setting?

The Approach

1. Loaded all three novels from the Gutenberg library as .txt files and built a word frequency function (get_wf) that strips punctuation and stop words, then counts occurring words remaining using Python’s Counter.

2. Extracted the top 15 most frequent words for each book, then plotted them side by side using horizontal bar charts (plotTwoLists) to compare.

3. Ran two comparisons: within Jane Austen’s two novels (*Pride and Prejudice vs Sense and Sensibility)* and between author (*Pride and Prejudice vs Marriage (Ferrier))*.

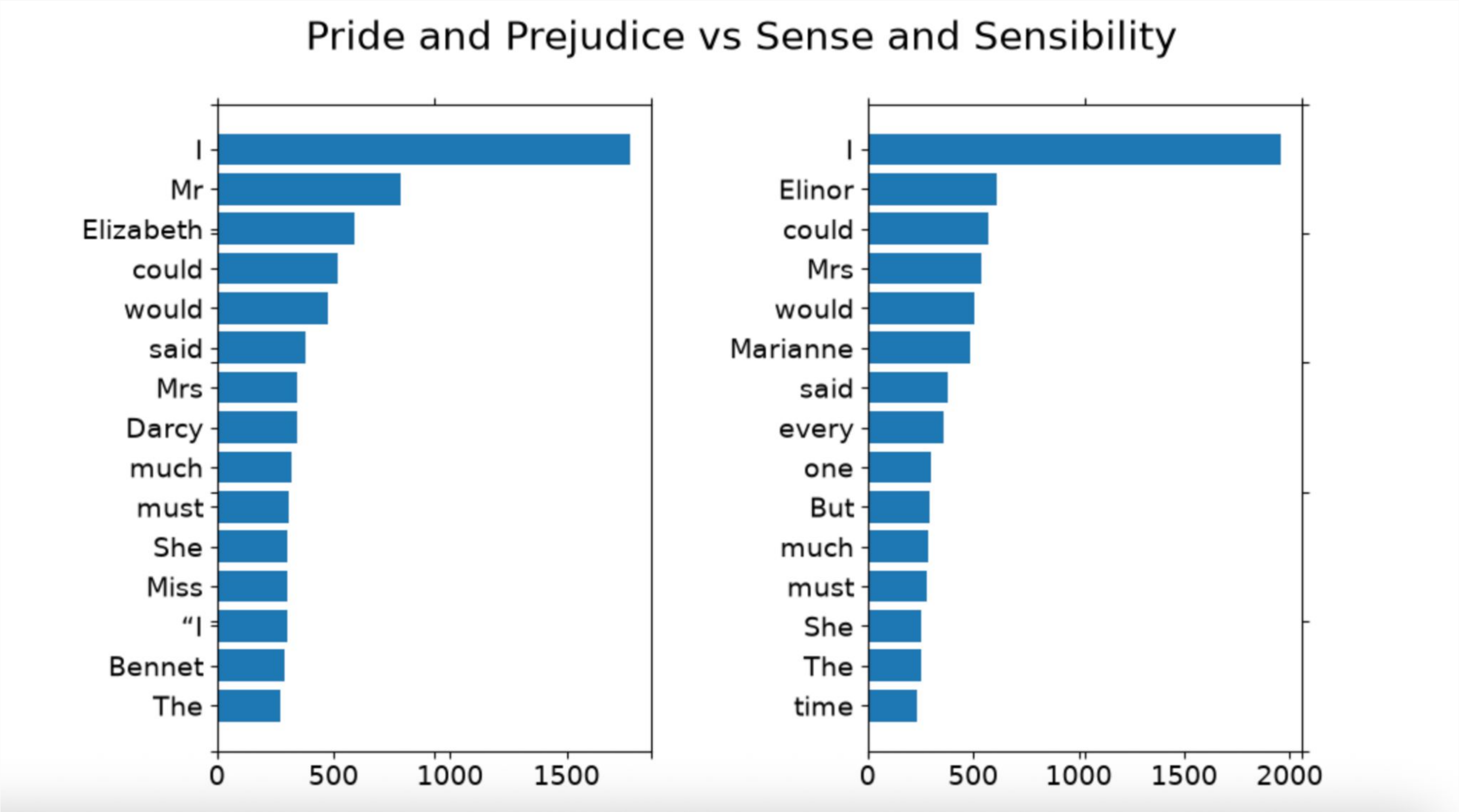
Problems Encountered

1. The matplotlib version mix-match caused a ValueError, which was fixed by removing the +1 in np.arange().

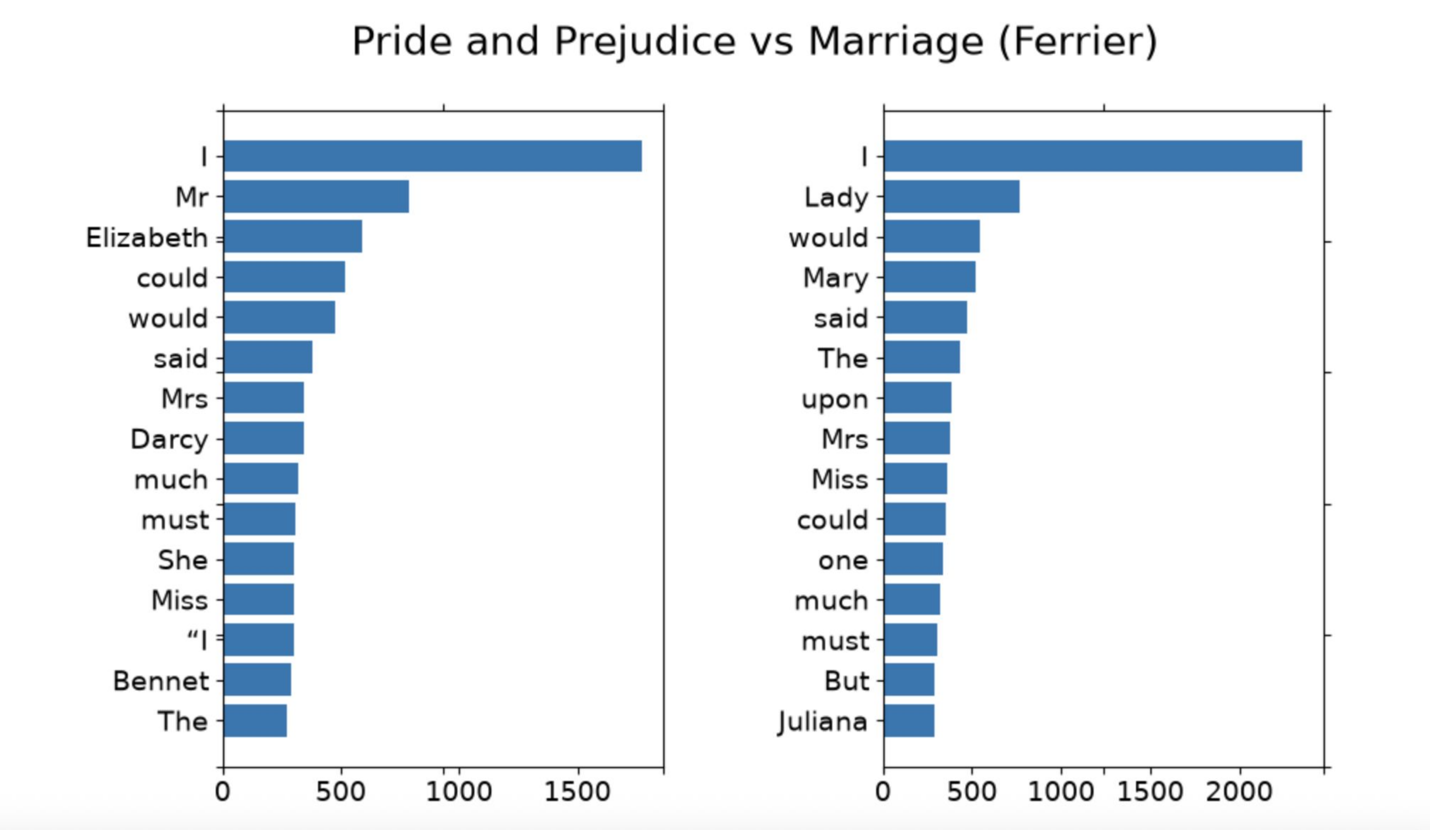
2. Another bug I discovered was indentetning that was simply fixed by moving plotTwoLists() outside of the function.

3. The top 15 word approach is dominated by function words (I, could, would, said, must, much, the) and character names (Elizabeth, Darcy, Bennet, Elinor, Mary, etc) rather than words that reveal style and phrasing.

Jane Austen Literature Results



Austen & Ferrier Comparison



Results

Within Austen:

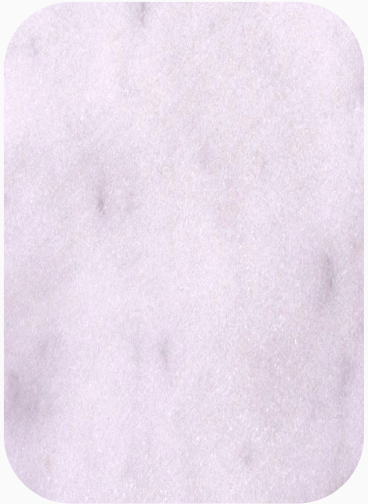
> The ranking and relative frequency of the function word is very similar across both books. This suggests that her core syntax and sentence structure vocab stayed stable across the two years between the books.

Austen vs Ferrier:

> Also more similar than expected at the function word level but “Lady” ranks very high in *Marriage* and “upon” appears in Ferrier’s top 15 but not Austen’s. This hints at Ferrier’s heavier uses of formal and aristocratic address and slightly more formal phasing.

Overall:

> The top 15 frequency view shows the two Austen novels are more alike than Austen to Ferrier, but the differences that show up are mostly proper nouns and titles rather than clear evidence of different language.



New Ideas Generated

- The top 15 view mostly surfaces character names and common functional words, it would help to look past these and compute a relative frequency difference between books (shown in extra credit).

- Excluding proper nouns and character names specifically could isolate the stylistic vocabulary from the character driven word counts.

- Looking for rarer words, average word length, or average sentence length across the books could speak more directly to “boldness” or the language differences.

- The extra credit test gives a better statistical answer to “did this really change” rather than an eyeballed visual comparison.

Extra Credit Results

```
=== Within Austen (PP vs SS) ===
Word: 'would'
  Pride and Prejudice: 477/65326 = 0.00730
  Sense and Sensibility: 509/61279 = 0.00831
  z = -2.032, p-value = 0.04218 --> SIGNIFICANT (p < 0.05)

Word: 'could'
  Pride and Prejudice: 519/65326 = 0.00794
  Sense and Sensibility: 569/61279 = 0.00929
  z = -2.583, p-value = 0.00980 --> SIGNIFICANT (p < 0.05)

Word: 'must'
  Pride and Prejudice: 309/65326 = 0.00473
  Sense and Sensibility: 283/61279 = 0.00462
  z = 0.292, p-value = 0.77054 --> not significant

Word: 'much'
  Pride and Prejudice: 322/65326 = 0.00493
  Sense and Sensibility: 284/61279 = 0.00463
  z = 0.759, p-value = 0.44788 --> not significant

Word: 'said'
  Pride and Prejudice: 380/65326 = 0.00582
  Sense and Sensibility: 381/61279 = 0.00622
  z = -0.921, p-value = 0.35689 --> not significant
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=== Austen vs Ferrier (PP vs Marriage) ===
Word: 'would'
  Pride and Prejudice: 477/65326 = 0.00730
  Marriage: 541/85797 = 0.00631
  z = 2.346, p-value = 0.01900 --> SIGNIFICANT (p < 0.05)

Word: 'could'
  Pride and Prejudice: 519/65326 = 0.00794
  Marriage: 350/85797 = 0.00408
  z = 9.845, p-value = 0.00000 --> SIGNIFICANT (p < 0.05)

Word: 'must'
  Pride and Prejudice: 309/65326 = 0.00473
  Marriage: 309/85797 = 0.00360
  z = 3.406, p-value = 0.00066 --> SIGNIFICANT (p < 0.05)

Word: 'much'
  Pride and Prejudice: 322/65326 = 0.00493
  Marriage: 324/85797 = 0.00378
  z = 3.403, p-value = 0.00067 --> SIGNIFICANT (p < 0.05)

Word: 'said'
  Pride and Prejudice: 380/65326 = 0.00582
  Marriage: 471/85797 = 0.00549
  z = 0.842, p-value = 0.39963 --> not significant
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Extra Credit

Question: Do these binomial distributions for your chosen word differ significantly between books of the same author or between authors?

The magnitude of difference is largest between the authors than within one authors own body of work. This gives backing to the visual pattern from the bar charts, that Austen's voice and tone are measurable more self-consistent than it it similar to Ferriers.

This test assumes independent word trials, which isn't really true of natural dialog and how phrases are structured.

Thank you